

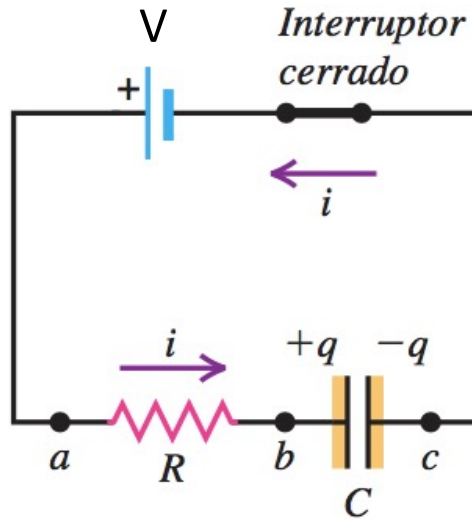
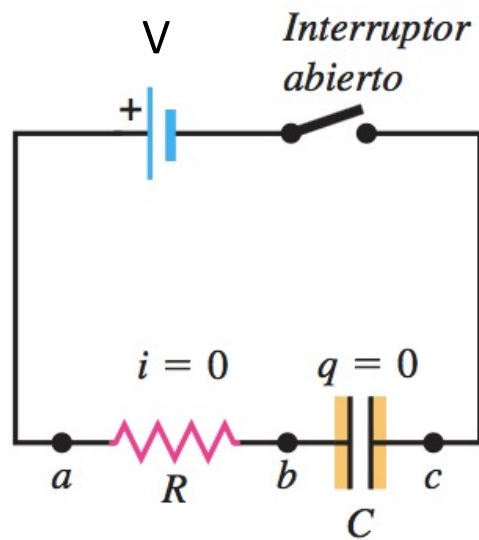
Clase 27

- circuito RC, circuito LR (repaso)
- circuito LC
- corrientes alternas
- reactancia

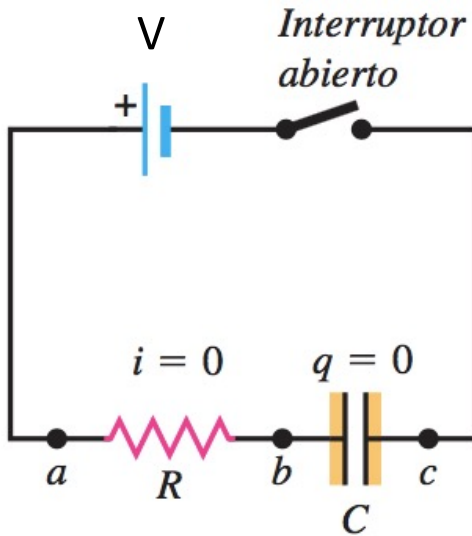
Sears CAP 31

Purcell CAP 8

Circuitos RC – carga de un condensador

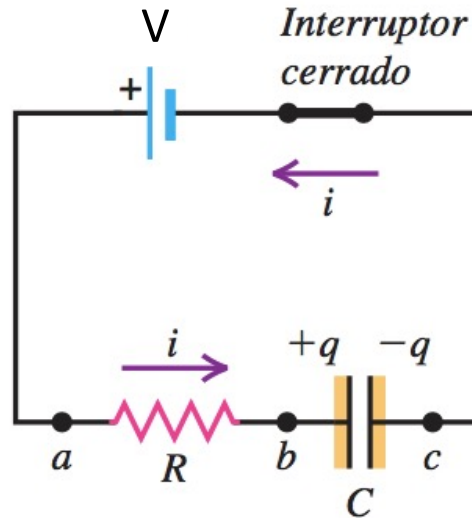


Circuitos RC – carga de un condensador

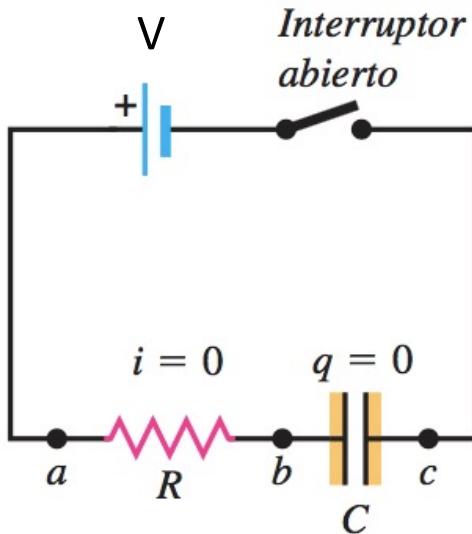


$$V - iR - V_c = 0 \rightarrow V - iR - \frac{q}{C} = 0 \rightarrow i = \frac{V}{R} - \frac{q}{CR}$$

$$\frac{dq}{dt} = \frac{V}{R} - \frac{q}{RC} = -\frac{q - CV}{RC}$$



Circuitos RC – carga de un condensador

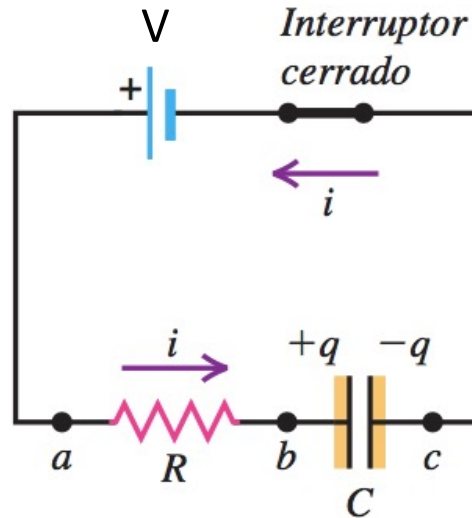


$$V - iR - V_c = 0 \rightarrow V - iR - \frac{q}{C} = 0 \rightarrow i = \frac{V}{R} - \frac{q}{CR}$$

$$\frac{dq}{dt} = \frac{V}{R} - \frac{q}{RC} = -\frac{q - CV}{RC}$$

$$\frac{dq}{q - CV} = -\frac{dt}{RC}$$

$$\int_0^q \frac{dq'}{q' - CV} = -\int_0^t \frac{dt}{RC} \quad \ln\left(\frac{q - CV}{-CV}\right) = -\frac{t}{RC} \rightarrow \frac{q - CV}{-CV} = e^{-t/RC}$$

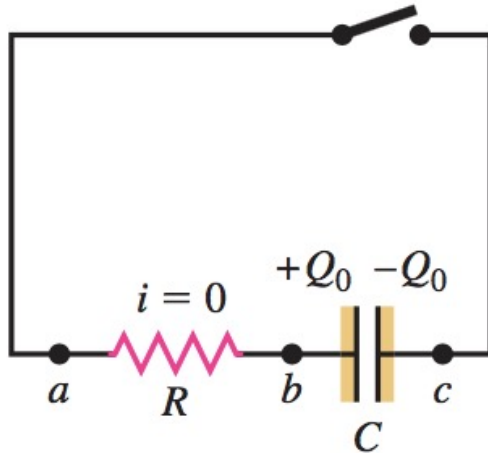


$$q(t) = CV(1 - e^{-t/RC})$$

$$i = \frac{dq}{dt} = \frac{V}{R} e^{-t/RC}$$

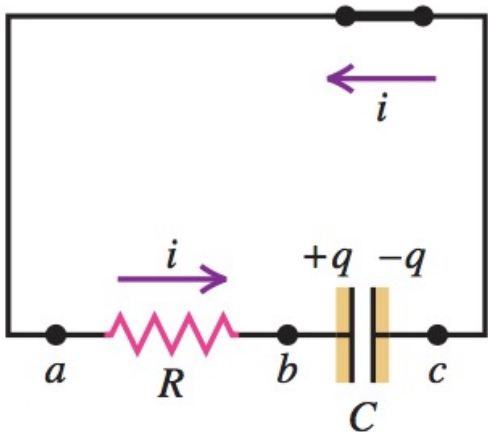
$$V(t) = \frac{q(t)}{C} = V(1 - e^{-t/RC})$$

Circuitos RC – descarga de un condensador



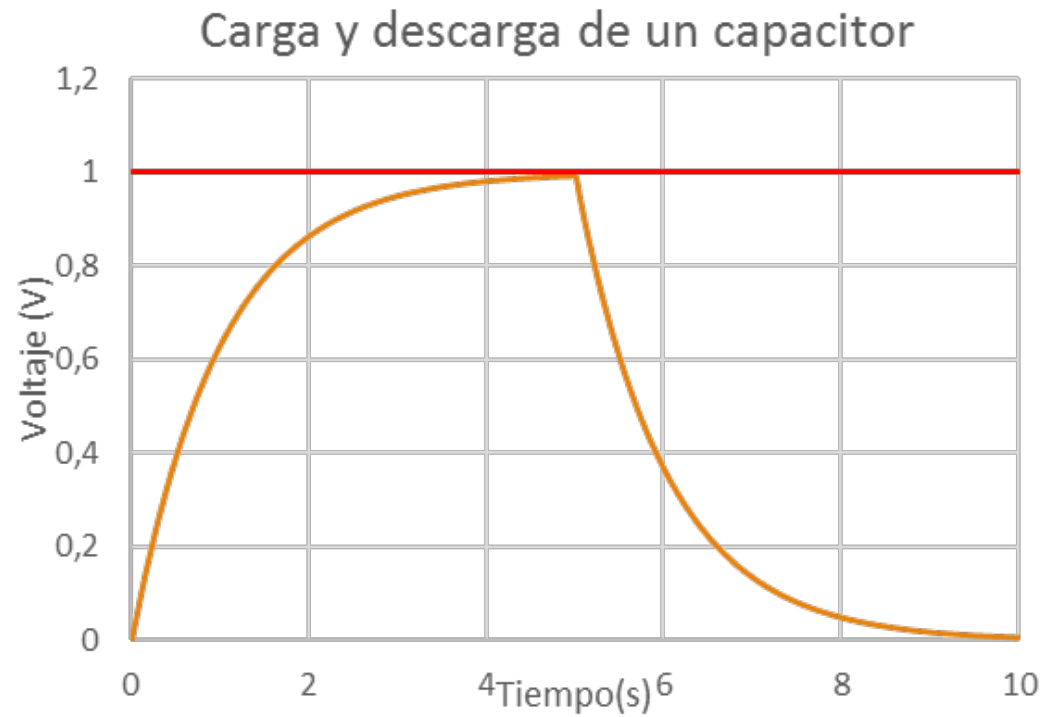
$$V \rightarrow 0 \quad \frac{dq}{dt} = \frac{V}{R} - \frac{q}{RC} \rightarrow \frac{dq}{dt} = -\frac{q}{RC}$$

$$\int_{Q_0}^q \frac{dq'}{q'} = -\int_0^t \frac{dt}{RC} \quad \ln\left(\frac{q}{Q_0}\right) = -\frac{t}{RC} \rightarrow q(t) = Q_0 e^{-t/RC}$$



$$i(t) = \frac{dq}{dt} = \frac{Q_0}{RC} e^{-t/RC}$$

$$V(t) = \frac{Q(t)}{C} = \frac{Q_0}{C} e^{-t/RC}$$



$$V(t) = V_0 \left(1 - e^{-t/RC}\right)$$

$$V(t) = \frac{q(t)}{C} = \frac{Q_0}{C} e^{-t/RC} \rightarrow V_0 e^{-t/RC}$$

$$q(t) = CV_0 \left(1 - e^{-t/RC}\right) \rightarrow Q_0 = CV_0$$

$$t = RC \rightarrow V = V_0 \left(1 - 1/e\right)$$

$$t = RC \rightarrow V = V_0 \frac{1}{e}$$

Circuitos LR

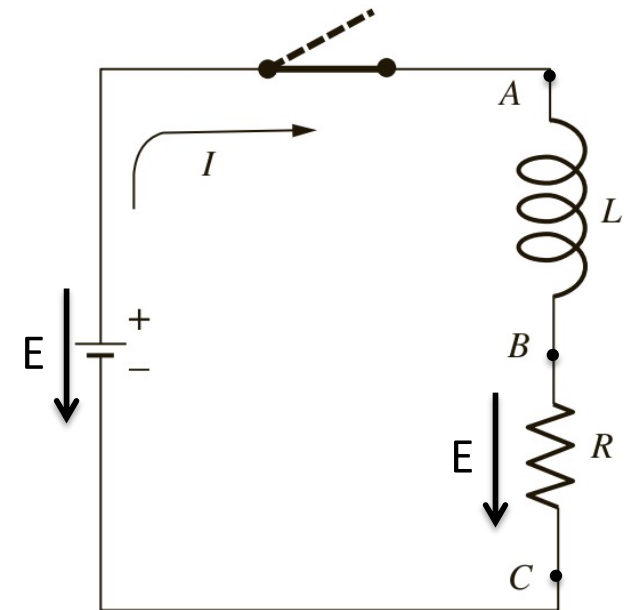
Un inductor se opone a los cambios de corriente y en régimen estacionario equivale a un cortocircuito

Faraday $\oint \vec{E} \cdot d\vec{l} = -\frac{\partial \Phi_B}{\partial t} = -L \frac{dI}{dt}$

$$0 + IR - V = -L \frac{dI}{dt}$$

$$IR = V - L \frac{dI}{dt}$$

La fem inducida se opone a V mientras que haya variación de corriente



por separación de variables

$$IR - V = -L \frac{dI}{dt} \quad \rightarrow \quad \frac{IR - V}{L} = -\frac{dI}{dt} \quad \rightarrow \quad -\frac{dt}{L} = \frac{dI}{IR - V}$$

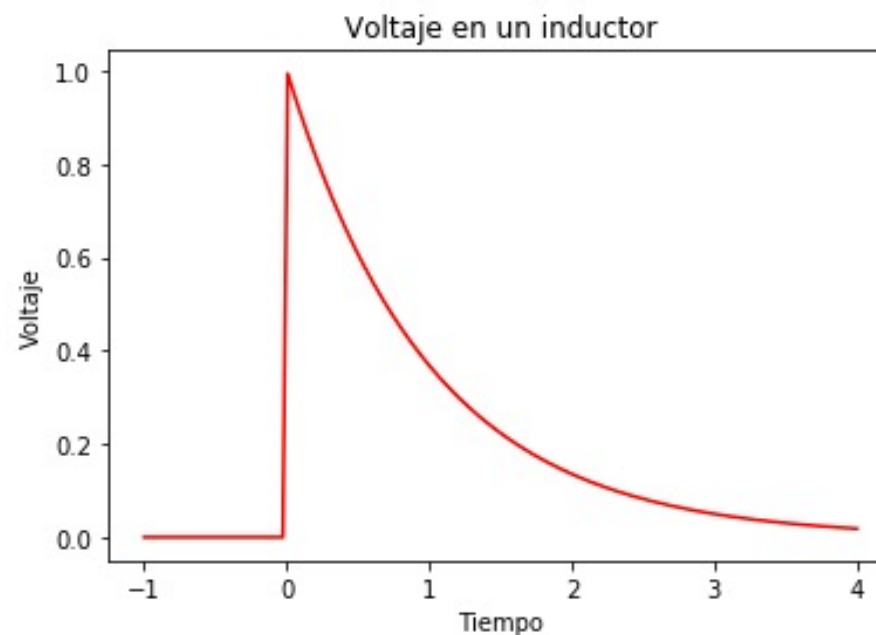
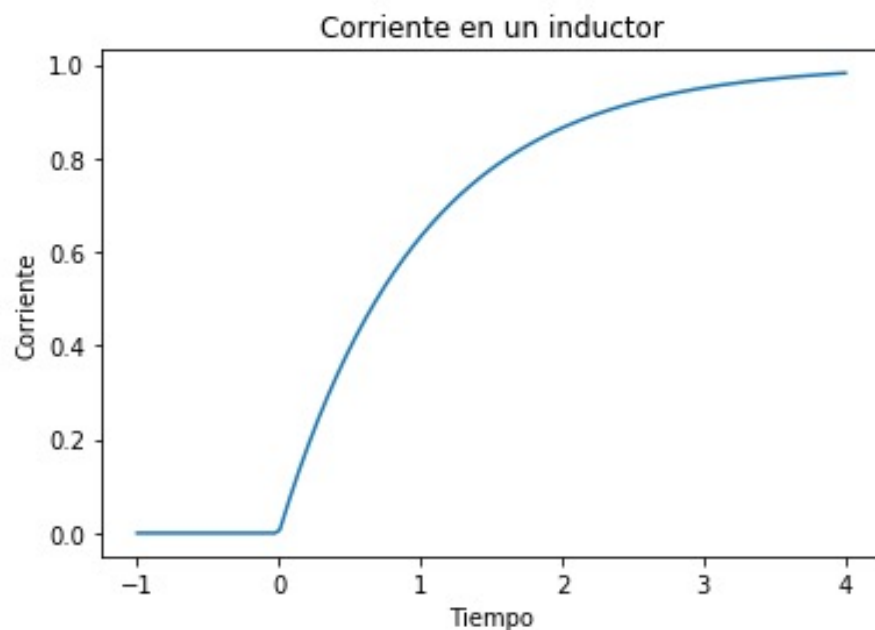
$$\frac{dI}{I - V/R} = -\frac{R}{L} dt \quad \rightarrow \quad \int_0^I \frac{di}{i - V/R} = -\frac{R}{L} \int_0^t dt$$

$$\ln(i - V/R)_0^I = -\frac{R}{L} t$$

$$\ln\left(\frac{I - V/R}{-V/R}\right) = -\frac{R}{L} t \quad \rightarrow \quad I(t) = \frac{V}{R} - \frac{V}{R} e^{-R/L t} = I_{\max} \left(1 - e^{-R/L t}\right)$$

$$E_{\text{ind}}(t) = -L \frac{dI}{dt} = V e^{-R/L t}$$

Voltaje y corriente en un inductor



Nótese que:

- Cuando se cierra interruptor, la oposición del inductor al flujo de corriente es ∞ (corriente 0)
- Para tiempo $\rightarrow \infty$, el voltaje en el inductor es 0 (es un corto-circuito) y la corriente tiende a V/R

$$V = 0 \Rightarrow IR - V = -L \frac{dI}{dt} \rightarrow IR = -L \frac{dI}{dt}$$

$$I(t) = I_{\max} e^{-R/L t}$$

energía disipada en ausencia del generador

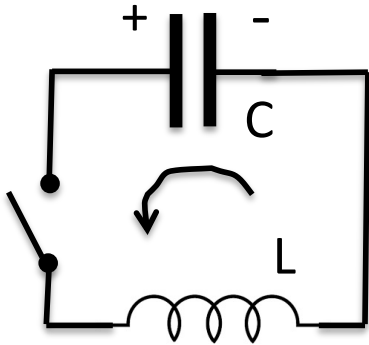
$$I^2 R \rightarrow \int_0^{\infty} I^2 R dt = I_{\max}^2 R \int_0^{\infty} e^{-2R/L t} dt = \frac{1}{2} I^2 L$$

alternativamente

$$dW = Vdq = L \frac{dI}{dt} (I dt) = LI dI \quad \int LI dI = \frac{1}{2} LI^2$$

lo que se puede imaginar parecido a la energía cinética $\frac{1}{2}mv^2$

Circuito LC



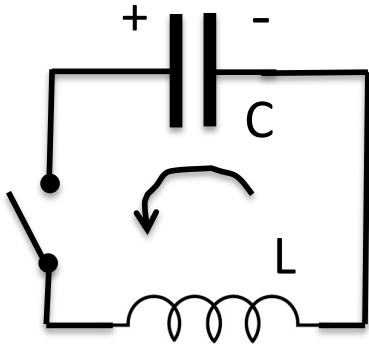
$$-V_c = -\frac{d\Phi_B}{dt}$$

$$-\frac{q}{C} = -L\frac{di}{dt}$$

$$i = -\frac{dq}{dt} \Rightarrow -\frac{q}{C} = L\frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} = -\frac{1}{LC}q$$

Circuito LC

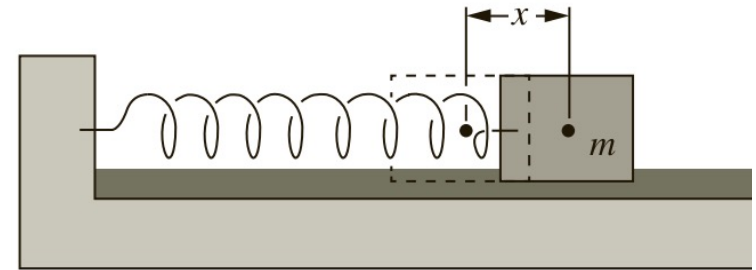


$$-V_c = -\frac{d\Phi_B}{dt}$$

$$-\frac{q}{C} = -L \frac{di}{dt}$$

$$i = -\frac{dq}{dt} \Rightarrow -\frac{q}{C} = L \frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} = -\frac{1}{LC}q$$



$$F = -Kx$$

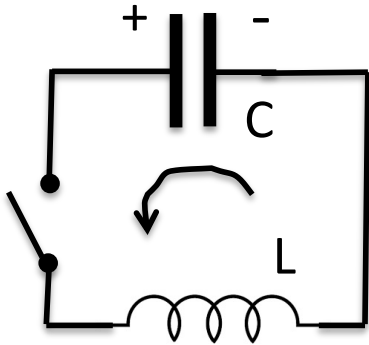
$$F = ma \Rightarrow -Kx = m \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} = -\frac{K}{m}x$$

$$x(t) = A \cos(\omega t)$$

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{K}{M}}$$

Circuito LC



$$-V_c = -\frac{d\Phi_B}{dt}$$

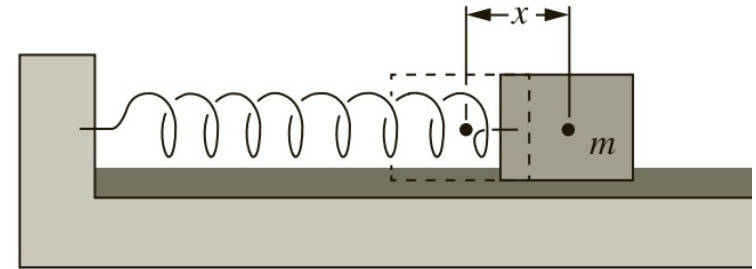
$$-\frac{q}{C} = -L \frac{di}{dt}$$

$$i = -\frac{dq}{dt} \Rightarrow -\frac{q}{C} = L \frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} = -\frac{1}{LC}q$$

$$q(t) = Q \cos(\omega t)$$

$$\omega = \sqrt{\frac{1}{LC}}$$



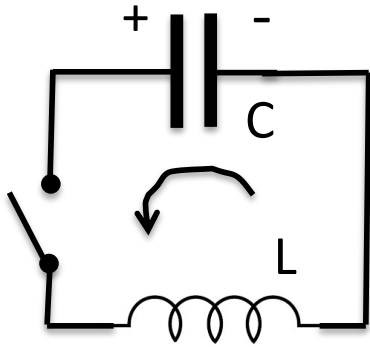
$$F = -Kx$$

$$F = ma \Rightarrow -Kx = m \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} = -\frac{K}{m}x$$

$$x(t) = A \cos(\omega t)$$

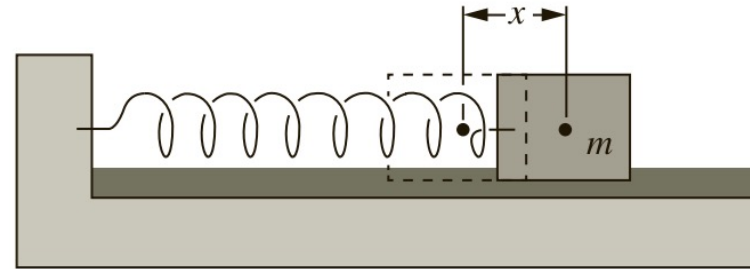
$$\omega = \frac{2\pi}{T} = \sqrt{\frac{K}{M}}$$



$$q(t) = Q \cos(\omega t)$$

$$\omega = \sqrt{\frac{1}{LC}}$$

$$E_{tot} = \frac{1}{2} L \dot{i}^2 + \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} \frac{Q^2}{C}$$



$$x(t) = A \cos(\omega t)$$

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{K}{M}}$$

$$E_{tot} = \frac{1}{2} m v^2 + \frac{1}{2} K x^2 = \frac{1}{2} K A^2$$

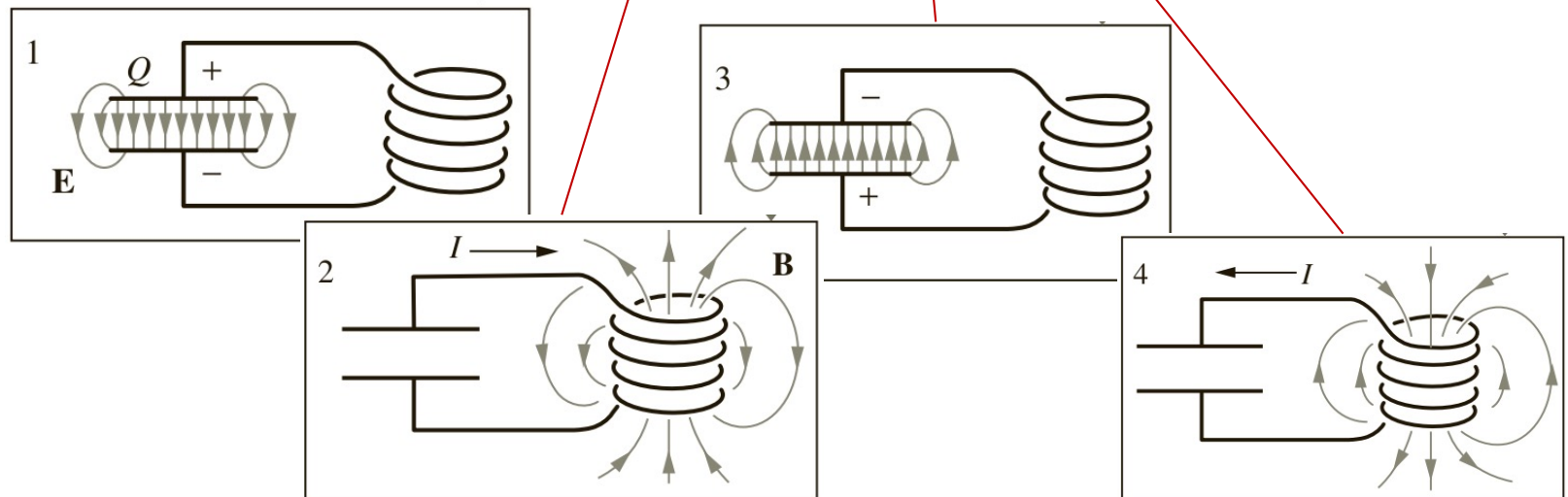
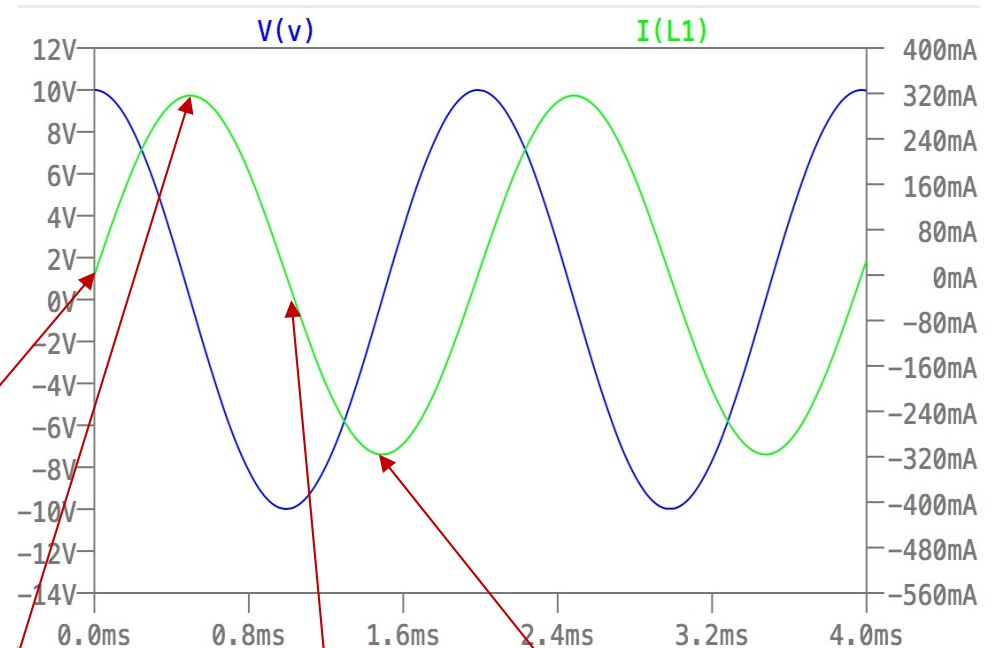
Circuito Resonante

Las cargas almacenadas en el capacitor generan corriente que es aceptada y almacenada por el inductor.

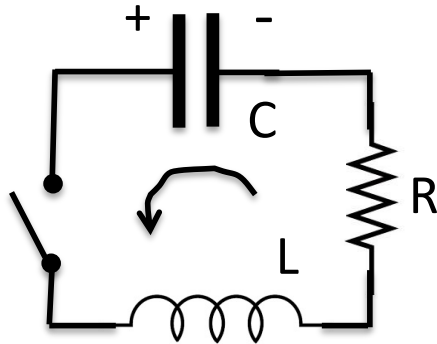
El inductor comienza a cargar el capacitor con polaridad opuesta.

Al no haber pérdidas, este proceso se alterna indefinidamente.

El análogo es un péndulo



Circuito LRC

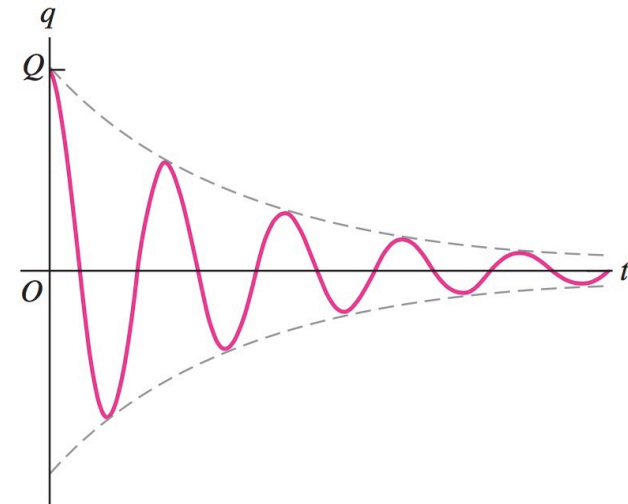


$$-V_c + iR = -\frac{d\Phi_B}{dt}$$

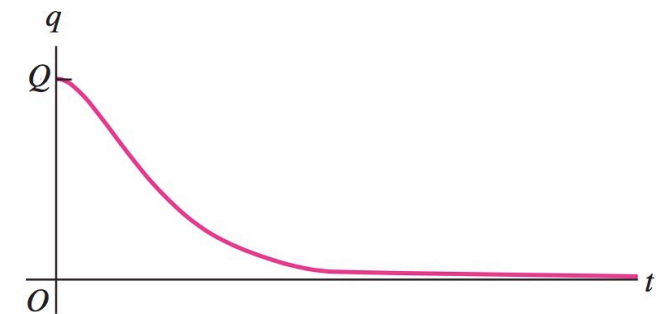
$$-\frac{q}{C} + iR = -L\frac{di}{dt}$$

$$i = -\frac{dq}{dt} \Rightarrow -\frac{q}{C} - \frac{1}{R}\frac{dq}{dt} = L\frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} + \frac{1}{LR}\frac{dq}{dt} + \frac{1}{LC}q = 0$$



sub-amortiguado $R^2 < 4L/C$



sobre-amortiguado $R^2 > 4L/C$

Corrientes Alternas

$$i = I \cos \omega t$$

$$v = V \cos \omega t$$

en Chile

$$f = 50 \text{ Hz} \Rightarrow \omega = 2\pi f = 377 \text{ rad/sec}$$

$$I_{rms} = \sqrt{\langle i^2 \rangle} \quad V_{rms} = \sqrt{\langle v^2 \rangle}$$

$$V_{rms} = \sqrt{\langle v^2 \rangle} = \sqrt{V^2 \langle \cos^2 \omega t \rangle} = \sqrt{V^2 \frac{1}{2}} = \frac{V}{\sqrt{2}}$$

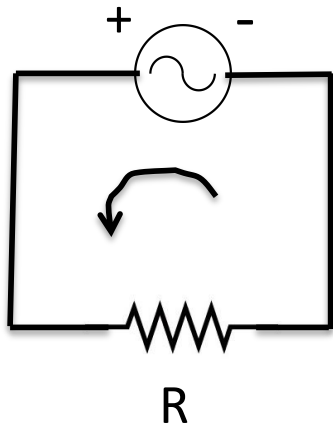
$$V_{rms} = 220 \text{ V} \Rightarrow V = 220\sqrt{2} = 311 \text{ V}$$

el famoso problema de los armónicos:

<http://www.fernandezfica.cl/armonicos.html>

<http://www.emb.cl/electroindustria/articulo.mvc?xid=3003&ni=la-creciente-problematica-de-los-armonicos>

Resistor en un circuito AC

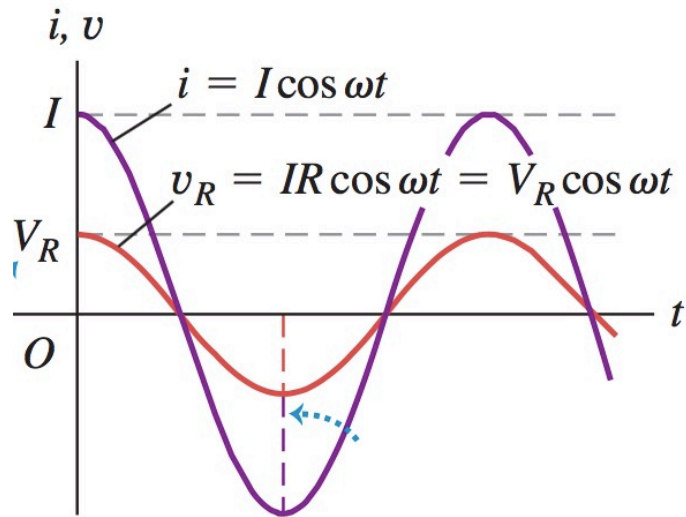


$$-v + iR = 0$$

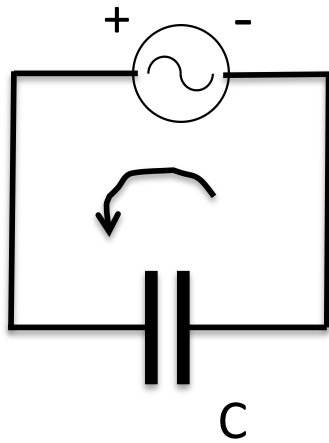
$$v = iR$$

$$i = I \cos \omega t \quad \Rightarrow \quad v = IR \cos \omega t = V \cos \omega t$$

$$V = IR$$



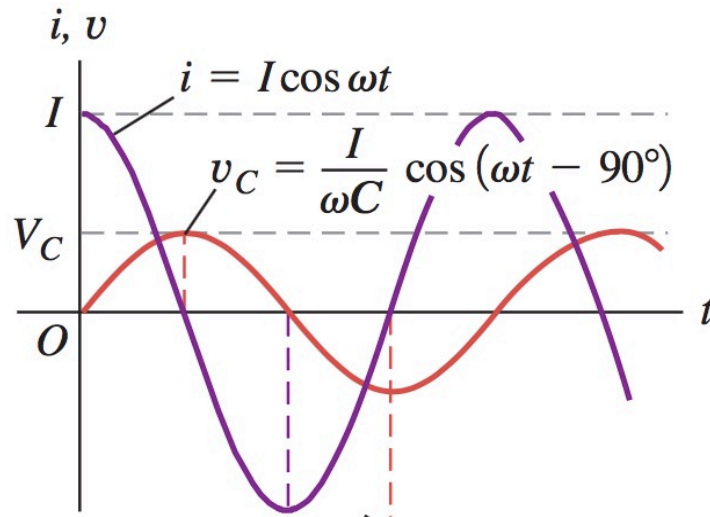
Capacitor en un circuito AC



$$-v + V_c = 0$$

$$v = V_c = \frac{q}{C}$$

$$i = \frac{dq}{dt} = I \cos \omega t \quad \Rightarrow \quad q(t) = \int I \cos \omega t \, dt = \frac{I}{\omega} \sin \omega t$$



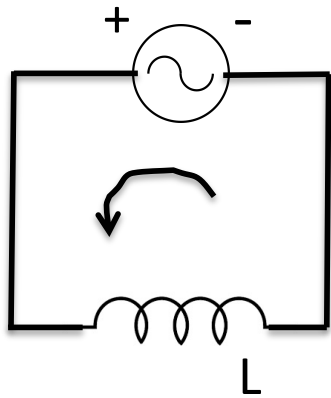
$$v(t) = \frac{q(t)}{C} = I \frac{1}{\omega C} \sin \omega t = I \frac{1}{\omega C} \cos(\omega t - 90^\circ)$$

reactancia
capacitiva

φ

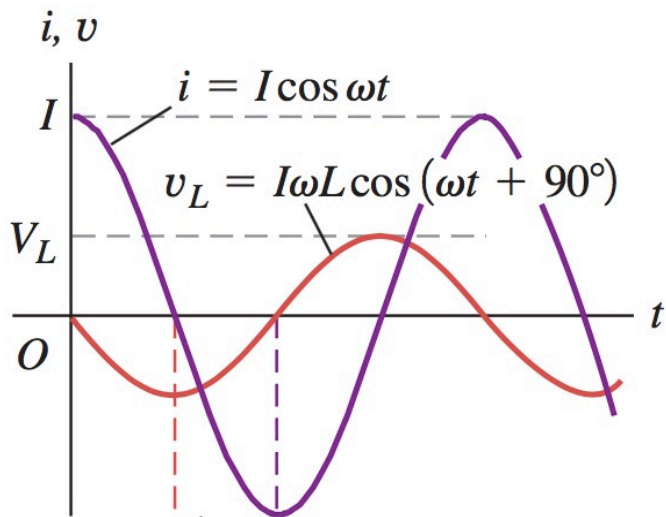
$$X_C = \frac{1}{\omega C} \quad \Rightarrow \quad V = IX_C$$

Inductor en un circuito AC



$$-v = -\frac{d\Phi}{dt} = -L \frac{di}{dt}$$

$$i = I \cos \omega t \Rightarrow v(t) = -LI\omega \sin \omega t = I \omega L \cos(\omega t + 90^\circ)$$



reactancia
inductiva

φ

$$X_L = \omega L$$

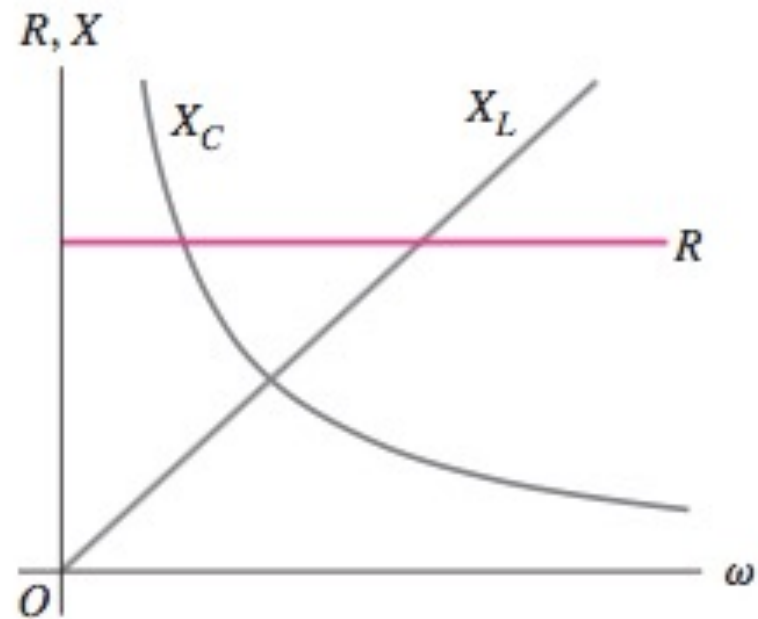
$$\Rightarrow V = IX_L$$

reactancia
inductiva

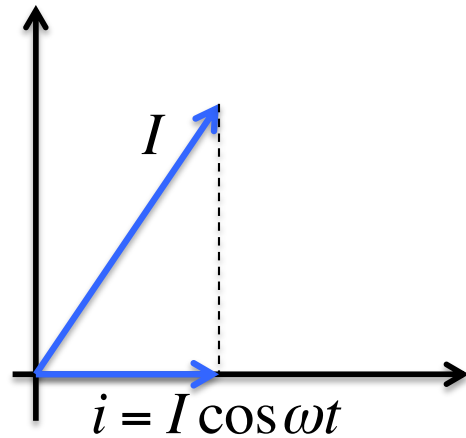
$$X_L = \omega L$$

reactancia
capacitiva

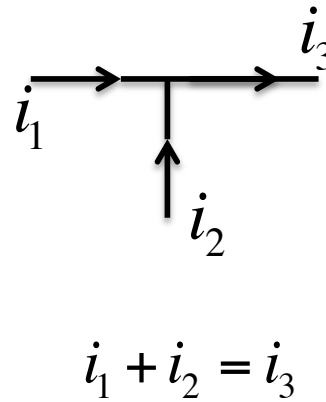
$$X_C = \frac{1}{\omega C}$$



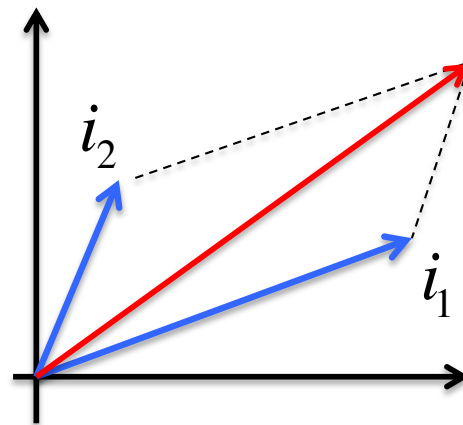
Diagramas de fasores



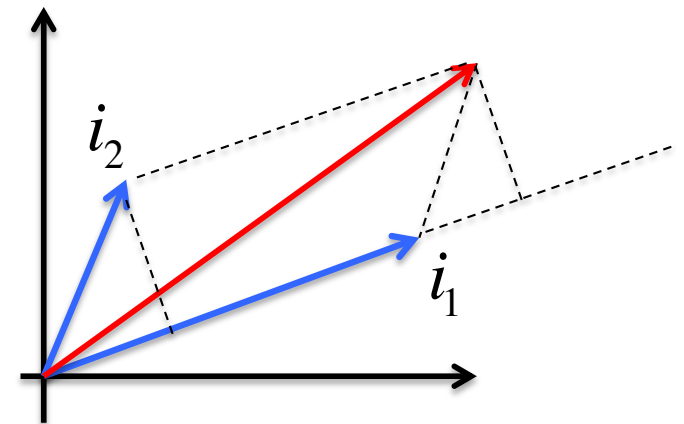
suma de corrientes con diagrama de fasores



$$\begin{cases} i_1 = i_{01} \cos \omega t \\ i_2 = i_{02} \cos(\omega t + \varphi) \\ i_3 = i_{03} \cos(\omega t + \psi) \end{cases}$$

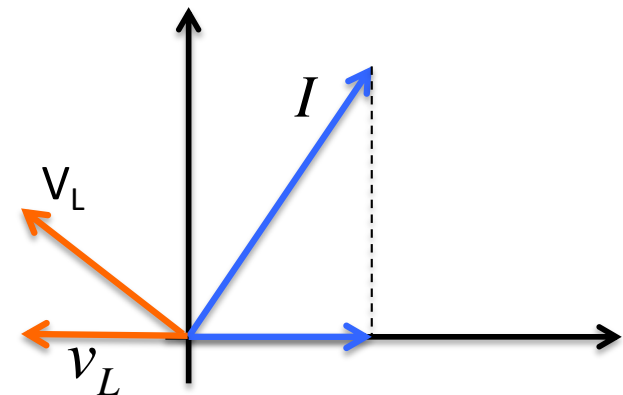
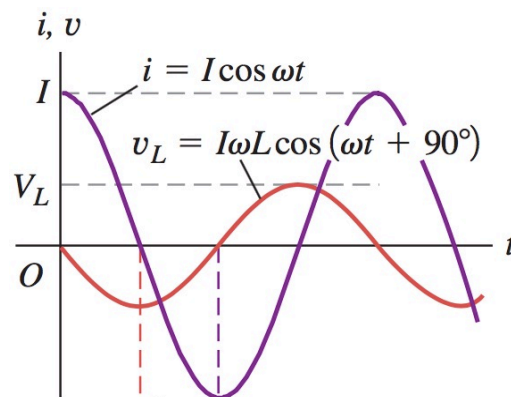
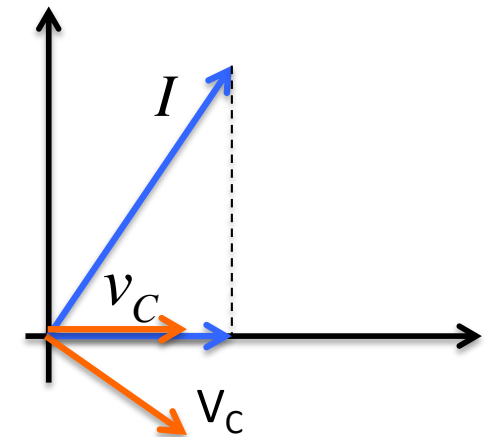
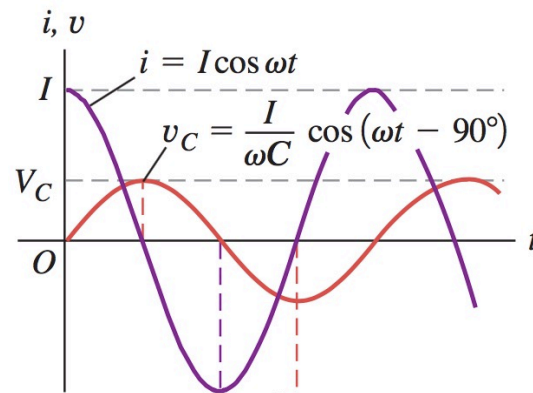
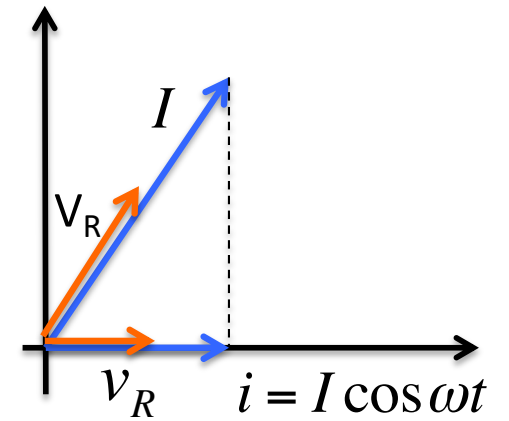
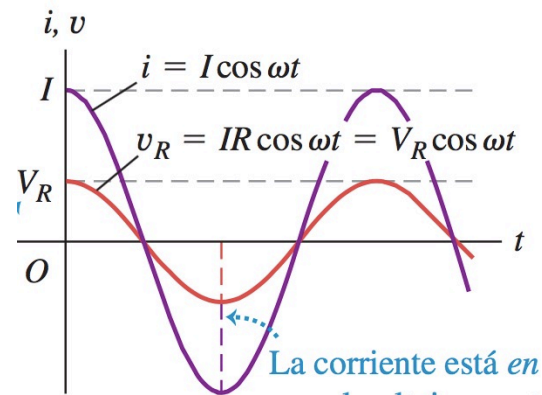
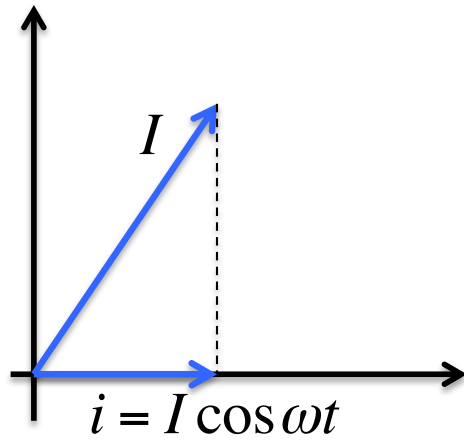


$$i_{03}^2 = i_{01}^2 + i_{02}^2 + 2i_{01}i_{02} \cos \varphi$$

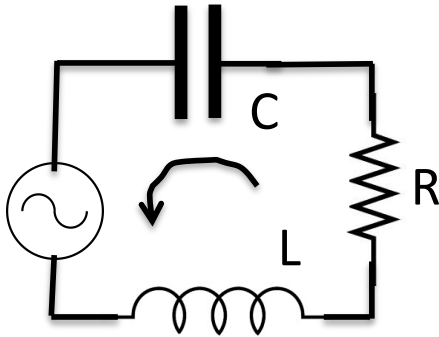


$$\tan \psi = \frac{i_{02} \sin \varphi}{i_{01} + i_{02} \cos \varphi}$$

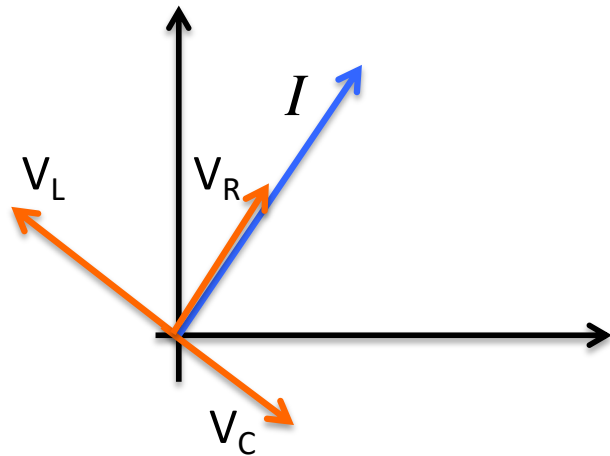
Diagramas de fasores



Circuito LRC



$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$$
$$= I\sqrt{R^2 + (X_L - X_C)^2} = IZ$$



$$Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2} \quad \text{impedancia}$$

$$\tan \varphi = \frac{V_L - V_C}{V_R} = \frac{\omega L - 1/\omega C}{R}$$

$$v = V \cos(\omega t + \varphi)$$

Circuito LRC en resonancia

$$V = IZ \Rightarrow I \text{ max cuando } Z \text{ min}$$

$$Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$$

condición de resonancia $X_L = X_C$

$$\omega L = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \quad I_{\text{max}} = V / R$$

$$\omega = 0 \Rightarrow Z = \infty \quad \text{por efecto de la capacitancia}$$

$$I = 0$$

$$\omega = \infty \Rightarrow Z = \infty \quad \text{por efecto de la inductancia}$$

$$I = 0$$

Purcell 8.2.5

